SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1515

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

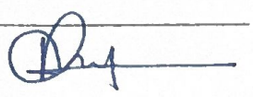
Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I VIGNESHWAR.D (192524415) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Concept Map 1 : cloud computing fundamentals.

Key characteristics

On-demand self-service
Users can provision resources automatically without human intervention.

Broad network access
Accessible over the internet from anywhere, anytime any device.

Resource pooling
Multiple users share the same physical resources.

Rapid elasticity
Resources can be scaled up or down quickly based on demand.

DEFINITION

"cloud computing is the on-demand delivery of computing resources over the internet."



LIMITATIONS :

- → Security + privacy risks.
- → Internet dependency.
- → Vendor lock-in.
- → Limited control.

BENEFITS :

Cost efficiency :
Reduces upfront hardware and maintenance costs.

Scalability :
Easily adjusts to changing demands.

High availability :
Ensures data and services are accessible.

Global reach :
Enables access from anywhere in the world.

Concept Map 2 : Evolution of cloud computing.

EVOLUTION OF CLOUD COMPUTING

The journey of computing from isolated hardware to internet-based, virtualized, and on-demand cloud services.

HARDWARE

→ Early computers were large, expensive and isolated.

→ Limited processing power & storage.

Personal computer

→ Affordable & compact system.

→ Individual users gained direct access to computing.

→ Increased personal.

Internet & web Application

→ Global connectivity through the internet.

→ Web browsers & online services.

→ Application became accessible from anywhere.

Distributed

→ Multiple computers work together over a network.

→ Better resource sharing & load balancing.

cloud Services

→ On-demand access to computing resources.

→ Pay-as-you go model.

→ Scalable, flexible & widely available.

MODERN CLOUD COMPUTING.

"A global, on-demand, Scalable & cost-effective model that delivers computers computing resources & services over the internet".

- GLOBAL ACCESSIBILITY
- SCALABILITY
- COST EFFICIENCY
- INNOVATION & GROWTH

Concept Map 3 : cloud Service Models.

cloud Service Models

"Provide different levels of abstration + management for building, running + scaling applications in the cloud"

DEFINED :

- → Level of control.
- → Managed Services.
- → user responsibilities.
- → Flexibility & Scalability.

TYPES :

- → IaaS
- → PaaS
- → SaaS
- → xaaS

Iaas
(Infrastructure as a Service)

provides virtualized computing resources over the internet. Includes servers, storage, networking + OS.

Paas
(Platform as a Service)

provide a complete development and runtime environment for building, testing + deploying.

Saas
(Software as a Service)

provides ready to use software application over the internet. Users access the software & a web browser.

xaaS
(Anything as a Service)

A broader term include any service delivered over the cloud, beyond the traditional Iaas.

FLEXIBILITY

Choose the right level of control and mangement as per need.

COST EFFICIENCY

Pay online for what you use + reduce infrastructure costs.

SCALABILITY :

Easily scale resources up or down based on demand.

INNOVATION :

Enables rapid development + access to advanced technologies.

Concept Map 4: Virtualization + web services as Enablers of cloud.

VIRTUALIZATION

→ A technology that creates as virtual version of physical hardware OS, storage, etc.

→ Allows multiple isolated environment (VMs) to run on a single physical machine.

- RESOURCE ABSTRACTION
- ISOLATION
- EFFICIENT UTILIZATION
- FLEXIBILITY + SCALABILITY

WEB - SERVICES

→ A set of self-contained, platform independent services that communicate over the internet.

→ Enables different applications + systems to interact and exchange data.

- STANDARDIZATION
- LOOSE COUPLING
- ACCESSIBILITY
- REUSABILITY

Cloud Deployment + Scalability

- Efficient Resource Management
- High Availability
- Elastic Scalability
- Global Access + Interoperability

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented